

Mean



1 Calculate the mean of the following data sets:

a 6, 4

b 8, 5

c 4, 8, 9, 12

d 7, 18, 9, 3

e 4, 10, 2, 9, 5

f 6, 14, 10, 13, 5, 9, 14, 15

g 11, 5, 14, 18, 15

h 8, 14, 24, 7, 12

i 13, 27, 9, 6, 8

j 13, 15, 7, 17, 9



2 State the mean of the following sets of data:

a 9, 9, 9, 9

b 2, 3, 4

c 850, 900, 950

3 Martha's grades in her 5 subjects are: 91, 93, 94, 90 and 92. Find the mean of Martha's grades.

4 Victoria has scored 12 goals this season, with a mean of 2 goals per game. How many games did she play?

5 James has played 10 games this season, with a mean of 1.5 goals per game. How many goals did he score in total?

Median



6 Find the median of the following data sets:

a 2, 3, 4, 5, 6

b 6, 9, 3, 5, 4

c 1, 4, 6, 8

d 5, 5, 5, 5

e 22, 12, 19, 16

f 2, 4, 9, 10

g 3, 18, 10, 19, 12, 5, 6, 20, 7

h 7, 19, 6, 14, 18, 11

i 5.3, 8.4, 2.6, 4.7, 3.5

j 49.4, 25.4, 47.2, 34.2, 28.4, 32.2

7 A set of 15 scores is arranged in increasing order. In what position does the median score lie?

8 A set of 23 scores is arranged in increasing order. In what position does the median score lie?

9 A set of 12 scores is arranged in increasing order. Between which two scores does the



- 10 A set of 34 scores is arranged in increasing order. Between which two scores does the median score lie?
- 11 Counting numbers in increasing order such as 4, 5, 6, 7 are called consecutive numbers. Write down five consecutive numbers whose median is 6.
- 12 Write down four consecutive odd numbers whose median is 12.
- 13 Write down four consecutive even numbers whose median is 15.
- 14 Find the median from the following frequency distribution tables:

a

Score	12	13	14	15	16
Frequency	1	2	3	4	5

b

Score	15	16	17	18	19
Frequency	1	4	1	4	4

c

Score	15	16	17	18	19
Frequency	2	7	3	1	5

d

Score	15	16	17	18	19
Frequency	1	2	5	5	2

e

Score	14	15	16	17	18
Frequency	1	2	5	5	1

f

Score	13	14	15	16	17
Frequency	5	4	1	3	5

Mode and range

- 15 Calculate the range of the following scores:

10, 16, 6, 18, 17, 11, 9, 15, 14

- 16 Write down four whole numbers whose median is 14 and whose range is 3.
- 17 Write down four whole numbers whose median is 17 and whose range is 3.

18 For each of the following data sets, calculate



- i The median ii The mode iii The range

a

Score	11	14	19	22	29
Frequency	10	13	4	7	12

b

Score	14	19	22	25	32
Frequency	9	14	12	1	10

19 A phone support centre tracks the length of calls made each day in minutes. During a shift one employee made calls of the following lengths:

2, 3, 3, 2, 2, 4, 4, 2, 5, 13

- a How many calls did the employee make during their shift?
b Calculate the following for the length of calls made:

- i Mean ii Median iii Mode iv Range

20 A student scored the following grades on his end of year tests:

87, 90, 85, 84, 89, 80, 85, 87, 87

- a How many tests did the student take?
b Calculate the following:

- i Mean ii Median iii Mode iv Range

21 A statistician organised a set of data into a frequency table:

Score	31	32	33	34	35
Frequency	12	14	7	20	15

- a Find the total number of respondents the statistician surveyed.
b Determine the following, rounding your answers to two decimal places:

- i Mean ii Median iii Mode iv Range

22 The television show "Dragon Duels" ran for 3 seasons, each with 6 episodes. The average fan ratings (out of 10) for each episode are shown in the following table:

	Episode 1	Episode 2	Episode 3	Episode 4	Episode 5	Episode 6
Season 1	8.2	8.4	7.9	7.9	8	8.2
Season 2	8.2	8.6	9.1	7.8	8.8	8.4
Season 3	8.9	9.4	9.2	9.7	7.4	4.3

- a** Which season has the highest mean rating?
- b** Which season has the highest median rating?
- c** Which season has the highest range of ratings?